Machine Learning

28-04-26

Overfitting:

- It occurs when train performance is very good but test performance is bad

causes:

1) Improper hyperparameter values
solution: Hyperparameter tuning

eg. DT max-depth = 20 is giving overfit model.
Tune max-depth by testing multiple lower max-depth values and find optimal value

for DTree models - as max-depth increases - chance of overfiting increases

2) using complex model for simpler task

model complexity ∝ overfiting

eg. using a deep neural network for linear regression task

using a random forest model with 10 decision trees for a simple binary classification

solution: use simple models in simple use-cases

3) Class imbalance issue

- model gets too much good on single dominant class and performs poor on rest of the classes

eg. target variable → 97% A, 2% B, 1% C

solutions:

1) improve data quality and reduce imbalance
2) oversampling minority classes
3) Apply SMOTE technique

SMOTE : Synthetic Minority Oversampling Technique

class imbalance issue: 97% A, $\underbrace{3\%\ B,\ 1\%\ C}$

- SMOTE tries to generate new synthetic datapoints for minority classes (B and C here)

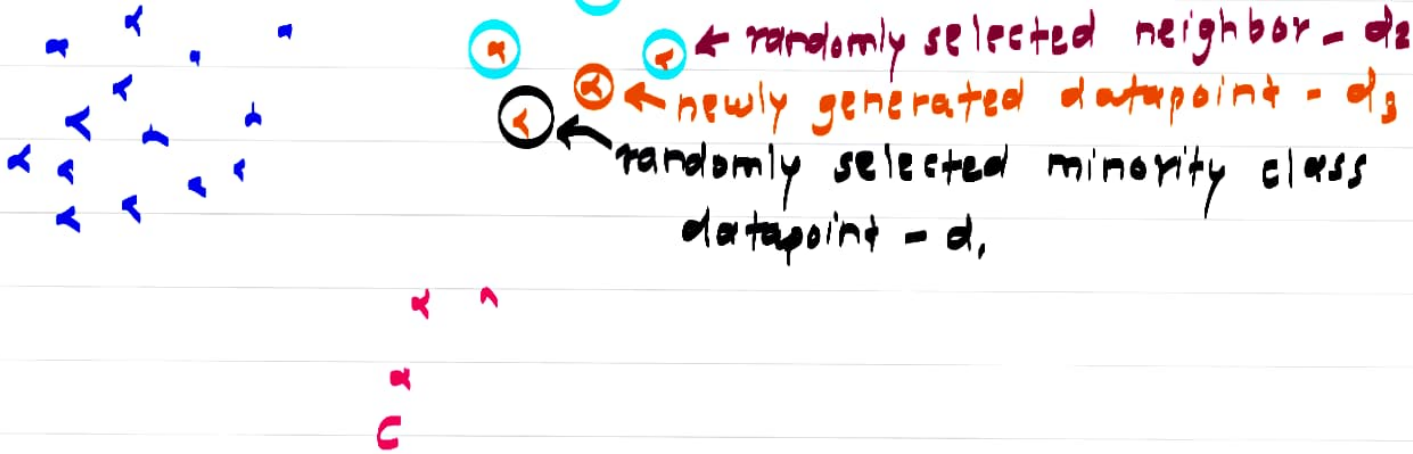
How SMOTE works?

1) Select a random datapoint from minority class

2) Find 'k' nearest neighbors for this datapoint within the same minority class

3) Select a random neighbor datapoint

4) generate a new datapoint in between neighbor and selected datapoint.



	x_1	x_2	target
d_1	10	5	B
d_2	4	8	B
d_3	7	6.5	B

$$d_3 = \frac{10+4}{2}, \frac{5+8}{2}$$

$$d_3 = 7, 6.5$$

we'll repeat this process and reduce class imbalance

4) Too many training features

- when train data has too many features, it creates a complex model causing overfiting

eg.

$$y = w_1 x_1 + w_2 x_2 + b$$ - simple model

$$y = w_1 x_1 + w_2 x_2 + w_3 x_3 \ldots\ldots w_{20} x_{20} + b$$

- complex model

more model parameters = more complex model

solution :

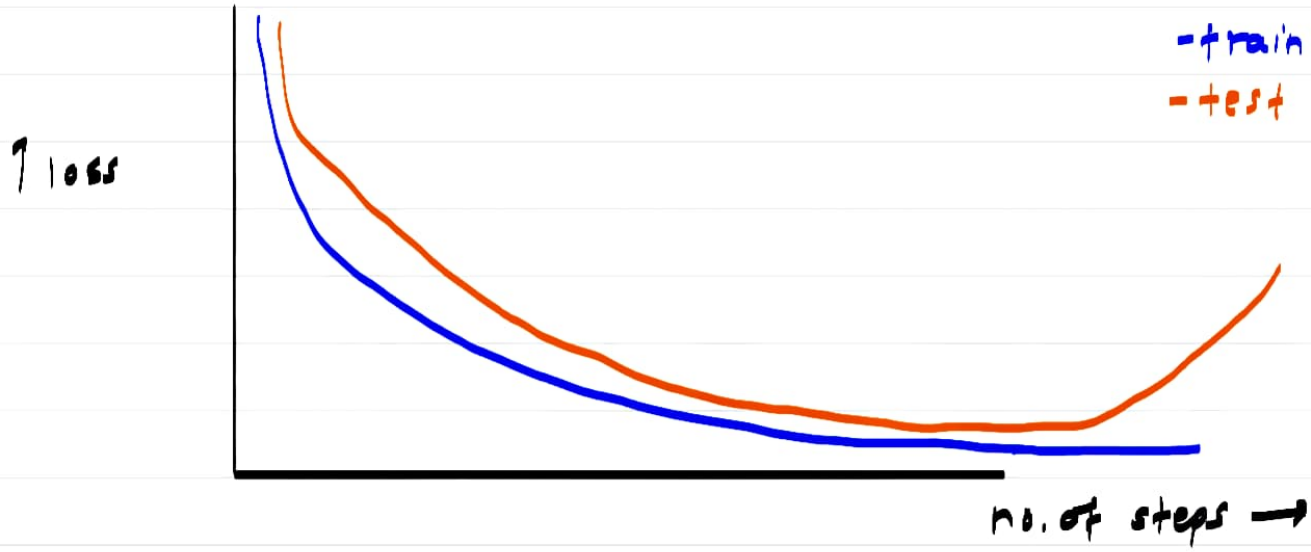
1) Feature engineering

- remove less important features
- apply variance thresholding

2) Regularization

- L1 / L2 regularization

5) Too many parameter updations in gradient descent



solution:

1) Apply early stopping

no. of steps	train loss	Validation loss test loss	model state
1	0.9	1.1	m_1
2	0.8	0.9	m_2
3	0.7	0.8	m_3
4	0.6	0.65	$\boxed{m_4}$
5	0.5	0.67	m_5
6	0.4	0.68	m_6
7	0.3	0.7	m_7

early stopping acts as a hyperparameter in model training

eg.

early stopping steps = 3

steps	train loss	val loss
1	0.9	1.2
2	0.8	1.1
3	0.7	1.05
4	0.6	0.95
5	0.5	0.98
6	0.45	0.99
7	0.4	0.97
8	0.38	0.96
9	0.35	0.95
10	0.31	0.96
11	0.3	0.965
12	0.28	0.97

val loss increased but only for 2 steps

loss increased for 3 steps

this will stop the model training

Underfiting:

- poor model performance on train and test data

causes:

1) Not choosing correct algorithm

- choose model according to use case.
- choose bigger and more complex model if simpler models are not working

Bias Variance Trade off:

underfit model has high bias
overfit mode has high variance

We need a best fit model which doesn't have high bias or high variance

The best fit model is also called 'generalised model' having moderate bias and variance

Regularization :

- its a technique to avoid overfiting of models
- most of the models have tendency to get overfit if not trained properly

Model	Regularization
- KNN	proper HP tuning for 'K'
- DT	proper HP tuning
- Linear / logistic regr	L1/L2 regularization
- Neural networks	L1/L2 regularization dropout, layernorm, batchnorm

L1 regularization → Lasso Regularization

L2 regularization → Ridge Regularization

both of these are useful in all gradient descent models.